A group is *residually finite* if every nonidentity element has a nonidentity image under a homomorphism to a finite group; equivalently, the intersection of the kernels of all such homomorphisms, the *finite residual*, is trivial. A finitely generated group is *word-hyperbolic* if some Cayley graph has uniformly thin geodesic triangles: a constant δ such that every point of one side of any geodesic triangle lies within δ of the other two sides. The residual finiteness question asks whether every word-hyperbolic group is residually finite, that is, whether coarse negative curvature by itself forces separation by finite quotients.
https://doi.org/10.1007/978-1-4613-9586-7_3

Hyperbolicity in this sense is Gromov's; the metric theory of polygonal and non-positively curved complexes, including the angular-link control of curvature used below, is developed by Bridson and Haefliger.
https://doi.org/10.1007/978-3-662-12494-9

Residual finiteness is common but far from automatic. Finitely generated free groups have it, by Stallings' finite-graph argument, and finitely generated linear groups over a field have it, by Malcev. Dehn's algorithm solves the word problem for many finitely presented groups, but solvability of the word problem does not by itself give residual finiteness, and the combinatorial background is in Lyndon and Schupp. At the opposite extreme, infinite simple groups such as Ol'shanskii's Tarski monsters admit no nontrivial finite quotient at all, so their finite residual is the whole group and they are emphatically not residually finite; none of these monsters is hyperbolic. Whether hyperbolicity rules this phenomenon out is the question.
https://doi.org/10.1007/978-3-642-61896-3
https://doi.org/10.1007/978-94-011-3618-1

Theorem 1. *There is a torsion-free word-hyperbolic group that is not residually finite.*

The group is the fundamental group $G = \pi_1(K, O)$ of a finite Euclidean triangle complex K. A finite family $\mathcal{R}$ of explicitly described nonidentity elements is produced, and no finite quotient separates all of them; consequently some fixed $w \in \mathcal{R}$ lies in every finite quotient, i.e. $w \in R_f(G)$, and since $w \neq 1$ the group fails to be residually finite. The argument proves that one member of $\mathcal{R}$ works for all finite quotients, without identifying which member.

The idea is short. The complex K is built from Euclidean triangles so that every vertex link has no short cycle; this much positive angle excess forces $\pi_1(K)$ to be torsion-free and word-hyperbolic. Inside K sit many four-edge loops, the *rectangular words*, each a nonidentity element. A single finite quotient can separate only finitely many elements, so residual finiteness would supply one finite quotient keeping all the rectangular words nontrivial at once. A rank count over a fixed finite field forbids this: each loop array contributes a matrix of large rank, while the local triangle relations force the same matrix to have small rank. Hence one rectangular word dies in every finite quotient.

The same obstruction has a consequence for linear representations. A group is linear over a commutative field k if it embeds in $\mathrm{GL}_n(k)$ for some n. Malcev proved that every finitely generated linear group over a commutative field is residually finite.
https://www.mathnet.ru/eng/sm6037

Corollary 1. *For the group G above, every commutative field k, every $n \geq 1$, and every homomorphism $\rho : G \to \mathrm{GL}_n(k)$, one has $R_f(G) \subseteq \ker \rho$. Thus one fixed nonidentity element is killed by every such representation, and G is linear over no commutative field.*

Proof. The image $\rho(G)$ is a finitely generated linear group, hence residually finite by Malcev. If $g \in R_f(G)$ had $\rho(g) \neq 1$, residual finiteness of $\rho(G)$ would give a finite quotient separating $\rho(g)$,

contradicting $g \in R_f(G)$. So $R_f(G) \subseteq \ker \rho$, and the nonidentity element of $R_f(G)$ is killed by every ρ. □

Several class-wide statements accompany the result. Kapovich and Wise proved that all hyperbolic groups are residually finite if and only if every nontrivial hyperbolic group has a nontrivial finite quotient, if and only if every hyperbolic group is virtually torsion-free. Agol, Groves, and Manning proved that a positive answer would make every quasiconvex subgroup of every hyperbolic group separable. Additional geometric structure does force residual finiteness: Haglund and Wise showed that compact virtually special cube complexes are linear, hence residually finite, and Agol showed that a word-hyperbolic group acting properly and cocompactly on a CAT(0) cube complex is virtually special, so a counterexample admits no such cubulation. Nonpositive curvature alone does not suffice: Wise's complete square complexes and Burger–Mozes' finitely presented simple groups act on products of trees and are not residually finite, though those spaces contain flat planes and are not hyperbolic. Nonlinearity also does not suffice: Kapovich produced a hyperbolic group all of whose finite-dimensional representations over any field have finite image, and Tholozan and Tsouvalis produced residually finite nonlinear hyperbolic groups.
https://doi.org/10.1006/jabr.1999.8104
https://msp.org/gt/2009/13-2/gt-v13-n2-p13-s.pdf
https://doi.org/10.1007/s00039-007-0629-4
https://doi.org/10.4171/DM/421
https://doi.org/10.4171/CMH/107
https://doi.org/10.1016/S0764-4442(97)86938-8
https://doi.org/10.2140/gt.2005.9.1915
https://doi.org/10.4171/CMH/581

The load-bearing input is the following criterion, whose hypotheses concern only the angular lengths of closed walks in vertex links.

A *finite Euclidean triangle complex* is a finite graph with finitely many Euclidean triangles attached side-to-side along full edges, with positive compatible lengths and isometric gluings. Each closed cell embeds; distinct cells may meet in a union of faces. The *angular link* at a vertex has one vertex per incident edge-germ and one edge per incident triangle corner, of length equal to the corner angle. A link walk is *reduced* if it has no immediate reversal, and *cyclically reduced* if this also holds across its closing seam; link distance between distinct components is $+\infty$. The *intrinsic metric* is the infimum of lengths of polygonal paths made of straight segments inside cells. A *turn* of a closed edge path compares the direction back along the incoming edge with the direction forward along the outgoing edge, as points of the vertex link.

Theorem 2 (Angular criterion)**.** *Let K be a connected finite Euclidean triangle complex. Suppose that for some $\eta > 0$ every nonempty cyclically reduced closed edge walk in every vertex link has angular length at least $2\pi + \eta$. Then $\pi_1(K)$ is torsion-free and word-hyperbolic. Moreover, any nonempty closed edge path whose every cyclic turn has link distance at least π is not nullhomotopic.*

The three conclusions are the closed-path statement, torsion-freeness, and word-hyperbolicity, proved after the construction. The criterion is the classical large-link curvature condition; the special case of metric polygonal complexes with angle $\leq 2\pi$ everywhere is Bridson–Haefliger's.

The combinatorial input is the high-girth alteration argument of Erdős.
https://doi.org/10.4153/CJM-1959-003-9

Fix $h = 20$, $c = 100^{-h}$, and an integer r with $cr \geq 100$; all limits are as the prime q tends to infinity, with these constants fixed. Put $P = \mathbb{F}_q^h$, $m = q^h$, and let $S \subset \mathbb{F}_q$ have $|S| = \lfloor q/100 \rfloor$.

A *mark* is a point of S^h. A *labelled line* is an affine line $\ell \subset P$ together with a marked point $a \in S^h \cap \ell$; write ℓ_i, a_i for label i, and $I(z)$ for the set of labels with $z \in \ell_i \setminus \{a_i\}$.

Lemma 1 (Incidence)**.** *For all sufficiently large primes q there are $N = (c+o(1))m$ labelled lines, indexed by I, and a set $E \subset P$ such that:*

(i) *the bipartite graph on $P \sqcup I$, with $z \sim i$ iff $z \in \ell_i \setminus \{a_i\}$, has no cycle of length ≤ 12;*

(ii) *$|I(z)| \ge 4r^2$ for $z \notin E$;*

(iii) *$|E| = o(m)$;*

and for these data $q > r$ and $\frac{Nr}{4} > 2m + r|E|$.

Proof. Begin with every point of S^h as a mark, each choosing its line direction independently and uniformly among the $J = (q^h - 1)/(q - 1)$ directions of lines through a point of P.

Degrees. For fixed z, each other mark lies on a line through z with probability $1/J$, so $|I(z)|$ is binomial with mean $\mu_z \sim cq$ uniformly in z and variance $\le \mu_z$. For large q one has $\mu_z \ge 8r^2$, and the second-moment inequality gives $\Pr(|I(z)| < 4r^2) \le \mu_z(\mu_z - 4r^2)^{-2} = O(q^{-1})$. Hence the expected number of low-degree points is $O(m/q)$, and by Markov's inequality, with probability tending to one, there are at most $mq^{-1/2}$ of them.

Short cycles. For $2 \le s \le 6$, count simple cycles with s points and s labels. There are at most m^s ordered choices of the points; consecutive points force a geometric line, on which the mark has at most q positions; and the direction of each of the s labels must equal the forced one, an event of probability J^{-s}, independent across labels (labels may even share a geometric line). So the expected number of such cycles is at most $m^s q^s J^{-s} \le q^{2s}$, since $J \ge q^{h-1}$, and the expected total over s is at most $6q^{12} \le q^{13}$ cycles of length ≤ 12.

Choose an outcome meeting both bounds. Delete the label of every line on a short cycle, at most $6q^{13}$ labels, and let E contain the surviving low-degree points and every point of a deleted line. Deleting labels creates no cycle and, off E, changes no degree, so

$$|E| \le mq^{-1/2} + 6q^{14} = o(m), \qquad N = (c + o(1))m.$$

As $cr \ge 100$, eventually $N/m \ge c/2$ and $r|E|/m \le 1$, while $cr/8 \ge 12.5 > 3$; thus $\frac{Nr}{4} > 2m + r|E|$. □

These data are fixed once and for all. No later finite quotient changes q or them.

Blocks. For $z \notin E$ put $d = |I(z)| \ge 4r^2$ and choose $n_2 \in \{r, \dots, 2r-1\}$ with $n_2 \equiv d \pmod r$, $n_1 = (d - (r+1)n_2)/r$; then $d = rn_1 + (r+1)n_2$ and $n_1 \ge (2r^2 - r + 1)/r \ge r$. Partition $I(z)$ into two *blocks* with $|I_b| = s_b t_b$ and $s_b, t_b \ge r$, one block having $(s_b, t_b) = (r, n_1)$ and the other $((r+1), n_2)$. Blocks at different points are indexed separately; $p(b)$ denotes the point of block b; points of E carry no block. Each block carries a bijection $i \mapsto (\alpha_b(i), \beta_b(i)) \in \mathbb{Z}/s_b\mathbb{Z} \times \mathbb{Z}/t_b\mathbb{Z}$.

The complex. Let $[r] = \{0, \dots, r-1\}$ and take vertices O, V, W with edges

$$x_{iuv} : V \to W \ (i \in I,\ u, v \in [r]), \quad L_{b,u,j} : O \to V \ (u \in [r],\ j \in \mathbb{Z}/s_b\mathbb{Z}), \quad R_{b,v,k} : O \to W \ (v \in [r],\ k \in \mathbb{Z}/t_b\mathbb{Z}).$$

For each block b, each $i \in I_b$, and all $u, v \in [r]$, attach one triangle with sides

$$x_{iuv}, \qquad L_{b,u,\alpha_b(i)+v}, \qquad R_{b,v,\beta_b(i)+u}, \tag{1}$$

and angles $3\pi/5$ at O, $\pi/5$ at V, $\pi/5$ at W, all of one fixed shape. Every triangle has three distinct vertices, so closed cells embed; unused edges get matching lengths. Regular points exist, so the complex K is connected, finite, with finitely many cells.

Lemma 2. *The complex K satisfies the angular criterion with $\eta = 2\pi/5$. At V or W the link distance between distinct x-directions of one label i is at least $14\pi/5$, or infinite.*

Proof. At O, a fixed block contributes a complete bipartite link on its L- and R-directions: for fixed u, v, j, k the equations $\alpha_b(i) = j - v$, $\beta_b(i) = k - u$ pick out exactly one $i \in I_b$. Distinct blocks are disjoint components, so every nonempty reduced closed walk has at least four edges, of angular length at least $12\pi/5$.

Project the link at V to the incidence graph by sending an x_{iuv}-direction to i and an $L_{b,u,j}$-direction to $p(b)$; this is locally injective. At x_{iuv} there is at most one incident triangle over any point z, since i lies in at most one block at z. At $L_{b,u,j}$ a fixed i determines at most one v from $j = \alpha_b(i) + v$, because $s_b \geq r$. The link at W behaves the same using $t_b \geq r$ and the R-directions. A cyclically reduced walk of length L in such a link maps to a walk of the incidence graph reduced at internal vertices; a shortest positive subwalk with equal endpoints is then a genuine cycle of length $\leq L$ and at least 14, so $L \geq 14$ and the angular length is at least $14\pi/5$.

A shortest link path between distinct x-directions with one label i is reduced; it projects to a positive closed walk at i reduced inside, which contains a cycle of length ≥ 14, so its angular length is at least $14\pi/5$ (infinite only if no path exists). □

Set $G = \pi_1(K, O)$. By the angular criterion and the link bound, G is torsion-free and word-hyperbolic; what remains is to exclude residual finiteness.

Rectangular words. Choose a path $O \to v$ for each vertex, giving an element g_e for each oriented edge, and write $g_{iuv} = g_{x_{iuv}}$. Each triangle (1) reads, in path order,

$$g_{iuv} = g_{L_{b,u,\alpha_b(i)+v}}^{-1} \, g_{R_{b,v,\beta_b(i)+u}} \qquad (i \in I_b). \tag{2}$$

Lemma 3. *For $i \in I$ and $u \neq u'$, $v \neq v'$ in $[r]$,*

$$w(i, u, u', v, v') := g_{iuv} \, g_{iu'v}^{-1} \, g_{iu'v'} \, g_{iuv'}^{-1} \neq 1. \tag{3}$$

Proof. After cancelling the connector paths, this word is conjugate to the closed four-edge loop $V \xrightarrow{x_{iuv}} W \xrightarrow{x_{iu'v}^{-1}} V \xrightarrow{x_{iu'v'}} W \xrightarrow{x_{iuv'}^{-1}} V$. At each of its four vertices the turn compares two distinct x-directions with the common label i (at W the pair $x_{iuv}, x_{iu'v}$; at V the pair $x_{iu'v}, x_{iu'v'}$; and so on). Every cyclic turn has link distance at least $14\pi/5 \geq \pi$, so the closed-path criterion makes the loop nontrivial. □

Let $\mathcal{R} \subset G$ be the finite set of the words (3); the lemma says $1 \notin \mathcal{R}$.

Rank over a finite field from rank over the reals. An integer matrix whose rows have Euclidean norm at most $\sqrt{r}$ loses at most a factor two in rank on reducing modulo a prime larger than r.

Lemma 4. *Let M be an integer matrix whose rows have Euclidean norm at most $\sqrt{r}$, with $r \geq 1$. For every prime $q > r$, $\operatorname{rank}_{\mathbb{F}_q} M \geq \frac{1}{2} \operatorname{rank}_{\mathbb{R}} M$.*

Proof. Let $d = \operatorname{rank}_{\mathbb{R}} M$; the case $d = 0$ is clear. A nonzero $d \times d$ minor B satisfies $1 \leq |\det B| \leq r^{d/2}$ by Hadamard. If $d' = \operatorname{rank}_{\mathbb{F}_q} B$, subtract integer lifts of d' spanning rows from the others; these determinant-preserving steps make $d - d'$ rows divisible by q, so $q^{d-d'} \mid \det B$ and $q^{d-d'} \leq r^{d/2}$. As $q > r$ this forces $d' \geq d/2$. □

Suppose, for a contradiction, that a finite quotient Q of G keeps every member of $\mathcal{R}$ non-identity. Put $D = |Q|$, and for $g \in G$ let $U(g)$ be the $D \times D$ regular permutation matrix of left

multiplication on Q (column vectors); U is a homomorphism with $U(g)^{\mathsf{T}} = U(g^{-1})$. For $i \in I$ form the block matrix

$$Y_i = \big(U(g_{iuv})\big)_{u,v\in[r]},$$

of size $rD \times rD$, with exactly r unit entries in each row, hence row norm $\sqrt{r}$.

Lemma 5. $\operatorname{rank}_{\mathbb{R}} Y_i \geq rD/2$ *and* $\operatorname{rank}_{\mathbb{F}_q} Y_i \geq rD/4$.

Proof. A nonidentity regular permutation has real trace 0 and the identity has trace D, so

$$\operatorname{tr}(Y_i Y_i^{\mathsf{T}}) = r^2 D, \qquad \operatorname{tr}\big((Y_i Y_i^{\mathsf{T}})^2\big) = (2r^3 - r^2)D.$$

Here $Y_i Y_i^{\mathsf{T}}$ has block $(u,u') = \sum_v U(g_{iuv} g_{iu'v}^{-1})$, and

$$\operatorname{tr}\big((Y_i Y_i^{\mathsf{T}})^2\big) = \sum_{u,u',v,v'} \operatorname{tr} U(g_{iuv} g_{iu'v}^{-1} g_{iu'v'} g_{iuv'}^{-1}).$$

The argument of $\operatorname{tr} U$ is trivial when $u = u'$ or $v = v'$ and is a word (3), hence nonidentity, otherwise; there are $2r^3 - r^2$ trivial index choices. Since $Y_i Y_i^{\mathsf{T}}$ is positive semidefinite, Cauchy–Schwarz on its eigenvalues gives

$$\operatorname{rank}_{\mathbb{R}} Y_i \geq \frac{(r^2 D)^2}{(2r^3 - r^2)D} = \frac{r^2 D}{2r-1} \geq \frac{rD}{2},$$

and Lemma 4 gives $\operatorname{rank}_{\mathbb{F}_q} Y_i \geq rD/4$. □

Work over $\mathbb{F}_q$ and put $Z(z) = \sum_{i\in I(z)} Y_i$.

Local factorization. Fix $z \notin E$ with its two blocks. For one block b,

$$\sum_{i\in I_b} Y_i = \Big(\sum_{j\in\mathbb{Z}/s_b\mathbb{Z}} U(g_{L_{b,u,j}}^{-1})\Big)_{u\in[r]} \Big(\sum_{k\in\mathbb{Z}/t_b\mathbb{Z}} U(g_{R_{b,v,k}})\Big)_{v\in[r]}, \tag{4}$$

a product of an $rD\times D$ block column and a $D\times rD$ block row. Its block (u,v) equals $\sum_{j,k} U(g_{L_{b,u,j}}^{-1} g_{R_{b,v,k}})$, which by the bijection $i \mapsto (\alpha_b(i)+v, \beta_b(i)+u)$ equals $\sum_{i\in I_b} U(g_{L_{b,u,\alpha_b(i)+v}}^{-1} g_{R_{b,v,\beta_b(i)+u}})$, i.e. $\sum_{i\in I_b} Y_i$ by (2). So each block has rank at most D, and

$$\operatorname{rank} Z(z) \leq \begin{cases} 2D, & z \notin E, \\ rD, & z \in E, \end{cases} \tag{5}$$

the second being the ambient size.

Interpolation. For $a_i = (a_{i1}, \ldots, a_{ih})$ put

$$f_i(X) = \prod_{t=1}^{h} \prod_{a\in S\setminus\{a_{it}\}} \frac{X_t - a}{a_{it} - a}, \qquad f_i(a_{i'}) = \delta_{ii'},$$

so $\deg f_i \leq h(|S|-1)$. Let $f = (f_i)_{i\in I}$ and $H(z) = f(z) f(z)^{\mathsf{T}}$, an $N \times N$ matrix of rank ≤ 1 with $H(a_i) = E_{ii}$. Each entry of H has degree $\leq 2h(|S|-1) < q-1$ and so, summed over the q points of any affine line, vanishes by the finite-field power-sum identity. Therefore

$$\sum_{z:\, i\in I(z)} H(z) = -H(a_i) = -E_{ii}, \tag{6}$$

and exchanging sums,

$$-\bigoplus_{i\in I} Y_i = -\sum_{i\in I} E_{ii}\otimes Y_i = \sum_{z\in P} H(z)\otimes Z(z). \tag{7}$$

Rank comparison. The left side of (7) has rank at least $\sum_i \operatorname{rank}_{\mathbb{F}_q} Y_i \geq NrD/4$. Each summand $H(z)\otimes Z(z)$ has rank at most $\operatorname{rank} Z(z)$, so summing over the m points and using (5),

$$\frac{NrD}{4} \leq 2D(m-|E|) + rD|E| = 2Dm + D|E|(r-2) \leq 2Dm + rD|E|.$$

Cancelling $D>0$ gives $\frac{Nr}{4} \leq 2m + r|E|$, contradicting the strict budget of the incidence lemma. Hence no finite quotient preserves all of $\mathcal{R}$.

If each $w\in\mathcal{R}$ had a nonidentity image in some finite group, the product of those finitely many maps, followed by taking its image, would be a finite quotient preserving all of $\mathcal{R}$. So there is one fixed $w\in\mathcal{R}$ whose image is the identity under every homomorphism to a finite group. Since $w\neq 1$ in G, the finite residual of G is nontrivial, and G is not residually finite. This proves the main theorem, subject only to the angular criterion, proved next.

The metric picture. Throughout, K is the complex above; "original" refers to its cells before subdivision, and links are metric graphs with edge lengths equal to corner angles. Barycentric subdivision gives a genuine simplicial complex S whose vertices are the original cells and whose simplices are strict inclusion chains (distinct parallel edges get distinct barycenters). Realized by straight segments, it preserves the link metric at original vertices; at a new vertex inside a triangle the link is a circle of length 2π; at a new vertex inside an edge the two tangent directions are joined by one path of length π per incident triangle (or are isolated). So every nonempty cyclically reduced closed link walk at a new vertex still has length at least 2π, and at an original vertex length at least $2\pi+\eta$. These facts persist under further compatible Euclidean subdivision, and the subdivision may be chosen to contain any fixed finite collection of edgeloops.

Lemma 6 (Metric preliminaries)**.** *The intrinsic distance is a metric inducing the complex topology; K is compact; any two points are joined by a shortest path; every constant-speed local geodesic, closed or not, has finitely many straight pieces. If such a path lies in the edges of a Euclidean subdivision, its link distance at each turn is at least π.*

Proof. Around x, the truncated incident sectors (and possibly isolated radial intervals) are glued along radial sides; the radial coordinate agrees across gluings, a path leaving the star of radius ρ has length $\geq\rho$, and the radial segment has length exactly ρ and is shortest. This gives positivity, identifies metric with star neighborhoods (so the metric induces the topology), and gives contracting neighborhoods. Compactness yields shortest paths as limits of proportional-arclength minimizing sequences: speeds are bounded, equicontinuity gives a uniformly convergent subsequence, and length is lower semicontinuous. A sufficiently short unit-speed minimizing arc inside a star has radial coordinate $r(t)=t$; its direction in the finite graph of directions is locally constant, so the arc is straight in one sector, giving finitely many straight pieces on any compact parameter set (and, for a closed geodesic, around the parameter circle). A turn whose angular distance is less than π unfolds two nearby equal-radius points into a sector of total angle $<\pi$ and a straight chord shorter than the two radii, contradicting local minimality; this includes the closing seam of a closed geodesic. □

Two-dimensional filling theory is classical; the account followed here is Lyndon and Schupp's.
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A *planar diagram* is a finite connected complex embedded in the plane, with edges and triangular faces, in which every bounded complementary face of its graph is filled by a triangle. Its *exterior walk* goes once around the unbounded face, counting incidences; it may repeat vertices and traverse a spur (an edge with no incident triangle) twice. The embedding fixes cyclic orders; the Euclidean metrics on the triangles need not come from this embedding.

Lemma 7 (Reduced fillings)**.** *Let S be a Euclidean simplicial subdivision of K. Every nonempty nullhomotopic closed edge path of S is the exterior walk of a planar diagram Δ carrying a map to S that takes each edge and triangle isometrically onto an edge and triangle. Every triangle has three distinct vertices and edges. At each interior vertex of Δ the cyclic sequence of corners maps to a nonempty cyclically reduced closed walk in the target link. Every edge loop in Δ contracts in Δ.*

Proof. Start from a continuous filling of the path by a disk, its boundary mapped linearly along each prescribed edge traversal, all boundary breakpoints marked. Barycentric coordinates of the vertices of S, extended by zero off their stars, are continuous; triangulate the disk finely enough that each coordinate varies by less than $1/4$ on each small triangle. Label a domain vertex by a target vertex of largest coordinate (at least $1/3$, as S has dimension ≤ 2), so at every point the chosen labels all have positive coordinate and lie in one simplex; extending linearly gives a simplicial map, allowing collapses. Along each prescribed boundary edge the only admissible labels are its two endpoints and the largest-coordinate label changes from one to the other exactly once, so the noncollapsed boundary steps are exactly the prescribed edge word in order; the same on a circle or interval makes every vertex loop a based edge-loop.

For the simplicial filling, draw the inverse images of the ties for the largest coordinate. In a triangle mapped onto a triangle these are three arcs meeting at a trivalent node; onto an edge, one separating arc; onto a vertex, absent. Suppressing valence-two bends gives a polygonal *picture*: regions are labelled by target vertices, an arc separates labels spanning a target edge, three labels at each node span a target triangle, and boundary leaves correspond to the letters of the edge word. Among all such pictures with the given boundary data choose one with the fewest nodes.

No arc joins two nodes labelled by one target triangle. If its two sides are a, b and d is the third vertex of that triangle, the arc with its endpoints is an embedded interval whose thin regular neighborhood meets the four other half-arcs in four distinct boundary slots; the two a-side slots both separate a from d, likewise the two b-side slots. Erasing the two nodes and the arc and rejoining each pair along its side, with label d between, gives a picture with two fewer nodes and unchanged boundary labels. This is unaffected by the half-arcs reconnecting outside the neighborhood. An arc cannot join a node to itself, since the three arcs there carry three distinct target-edge labels.

Next erase every component off the boundary. The retained picture with the boundary circle is a connected polygonal graph H whose inside regions are disk interiors with (possibly repeated) boundary walks. A thin regular neighborhood of H, missing the finitely many erased compacta, carries in each region a connected curve following the whole boundary walk on the region side (both sides of a bridge, every incidence at a repeated vertex); this curve crosses no old arc, so all boundary labels of the region agree, and the label extends over it. Erasing cannot add nodes, so minimality survives.

Now dualize: a vertex per region, radii to arc midpoints joined across arcs as dual edges; around each node the three radii bound a triangle whose dual vertices are distinct. With E arcs, N nodes, $L > 0$ boundary leaves, adjoining the leaf-subdivided circle gives a connected planar graph with $N + L$ vertices, $E + L$ edges, and $E - N + 1$ inside regions; the dual has $E - N + 1$

vertices and E edges, hence exactly N bounded faces, exhausted by the N constructed triangles. This is the required holed-free planar diagram mapping to S, its exterior walk the prescribed word and a boundary-to-boundary arc a spur-edge traversed twice. At an interior dual vertex an immediate link reversal would put two adjacent triangles on one target triangle, joined by the arc dual to their shared edge, contradicting minimality, cyclic seam included. Every simple edge circuit bounds its filled disk, and any edge loop splits into circuits and cancellable backtracks, so edge loops contract. □

For a diagram Δ and a vertex y, let θ_y be its total angle and n_y its number of exterior visits. Then $\theta_y \geq 2\pi$ when $n_y = 0$, and $\theta_y \geq 2\pi + \eta$ when also y maps to an original vertex of K. Angular Gauss–Bonnet is the identity

$$\sum_y \big((2 - n_y)\pi - \theta_y\big) = 2\pi, \tag{8}$$

the sum over all vertices of Δ.

Proposition 1 (Closed paths)**.** *In K, or any compatible Euclidean simplicial subdivision, a nonempty closed edge path whose every cyclic turn has link distance at least π is not nullhomotopic.*

Proof. Work in a simplicial subdivision and suppose such a path nullhomotopic, so Lemma 7 supplies a reduced diagram. At an interior vertex the summand of (8) is at most 0. At a vertex with $n_y = 1$ the two interior corners give a link path of length θ_y between the two boundary directions, so the prescribed turn forces $\theta_y \geq \pi$ and the summand is at most 0; the only cornerless case is a spur tip with coincident directions, excluded. For $n_y \geq 2$ the summand is at most 0 outright. All summands nonpositive contradict (8). A path in the original complex passes to the barycentric subdivision with unchanged turn distances (straight-through directions along an original edge have link distance π, or are an excluded spur), preserving the hypothesis. □

Shortest free loops. A closed local geodesic bounds a free homotopy class in which it is shortest; the closed-path obstruction then kills torsion.

Proposition 2 (Torsion-free)**.** *$\pi_1(K)$ is torsion-free.*

Proof. Contracting stars and compactness give $\epsilon > 0$ so every ball of radius 5ϵ is contractible in K, and any two loops uniformly ϵ-close are freely homotopic: cone each quadrilateral through shortest connectors in a ball of radius 3ϵ and glue along the connectors into an annulus. Let $g \neq 1$ and take the infimum $\ell \geq 5\epsilon$ of lengths in its free class. A proportional-arclength minimizing sequence has bounded speed, so Lemma 6 gives a length-ℓ minimizer in the same class; arclength parametrization shows it is a local geodesic, since otherwise a subarc of length $< \epsilon$ is replaced by a shorter shortest path with the same endpoints, the difference contracting in a star and shortening the loop within its class. This closed local geodesic has finitely many straight pieces; insert them into a finite compatible Euclidean subdivision so it lies in the edges, preserving links and turn distances. If g had finite order, a positive repetition would be nullhomotopic, contradicting Proposition 1, since its turns, including repeated seams, still have link distance at least π. □

Uniform thinness. From here S is the fixed barycentric subdivision of K, so every small triangle has an original vertex among its vertices. A reduced diagram over S has total angle at least 2π at every interior vertex and at least $2\pi + \eta$ at one mapping to an original vertex.

Lemma 8 (Thin diagrams)**.** *Let M exceed every diameter and edge length among the finitely many model triangles and edges, and put $A = M + 2M\pi/\eta$, $\delta = 10(A + 1)$. Every geodesic triangle in every reduced diagram is δ-thin.*

Proof. Fix a point p of one side off the other two; let (a, b) be the component of this side off the union of the other two that contains p. A simple path Q on the other sides runs from a to b in at most two geodesic subarcs (a subarc if one side holds both, else the first meeting of the two arcs toward the common corner). Then $[a, b] \cup Q$ is a simple closed curve of at most three geodesic sides and bounds a disk Ω, since diagrams have no holes; insert the sides' finitely many straight pieces and triangulate.

Let n count vertices of Ω° mapping to original vertices. Each has full excess $\geq \eta$; other interior vertices have angle $\geq 2\pi$; boundary vertices other than the at most three corners have interior angle $\geq \pi$ (else a local shortcut); corners have deficit $\leq \pi$. Lemma 7 with angular Gauss–Bonnet on the disk gives $2\pi \leq 3\pi - n\eta$, so $n \leq \pi/\eta$. For the inradius: for interior x join x inside a small triangle to an original vertex, length $< M$; either this meets $\partial\Omega$, or ends at one of the n counted vertices. So $\{d_\Delta(\cdot, \partial\Omega) > M\}$ lies in n ambient balls of radius M; as the distance is 1-Lipschitz and continuous on the compact connected Ω, it realizes $(M, \max]$ within total length $\leq 2Mn$, giving $d_\Delta(x, \partial\Omega) \leq A$ for all $x \in \Omega$.

Suppose $R := d_\Delta(p, Q) > 10(A+1)$. Interpolate $d_\Delta(p, \cdot)$ linearly on a fine subdivision of Ω to h with $|h - d_\Delta(p, \cdot)| < 1$, exact on the geodesic $[a, b]$, and $\geq R-1$ on Q. Pick a level $t \in (R/3, 2R/3)$ avoiding vertex values. Since $d_\Delta(p, a), d_\Delta(p, b) \geq R$, the level meets the boundary only on $[a, b]$, once on each side of p; its component is an arc γ with both ends on $[a, b]$. For $y \in \gamma$: $R/3 - 1 < d_\Delta(p, y) < 2R/3 + 1$ and $d_\Delta(y, Q) > R/3 - 1 > A$, so a nearest boundary point to y lies on $[a, b]$ at distance $\leq A$. As y runs over γ, $d_\Delta(a, y) - d_\Delta(a, p)$ changes sign, so some y has $d_\Delta(a, y) = d_\Delta(a, p)$; pick $z \in [a, b]$ with $d_\Delta(y, z) \leq A$. Geodesicity of $[a, b]$ gives $d_\Delta(z, p) = |d_\Delta(a, z) - d_\Delta(a, y)| \leq A$, whence $d_\Delta(p, y) \leq 2A$, contradicting $d_\Delta(p, y) > R/3 - 1 > 2A$. So $d_\Delta(p, Q) \leq 10(A + 1)$. □

Lemma 9 (Stability of controlled paths)**.** *Let X have δ-thin geodesic triangles, $\lambda \geq 1$, $C \geq 0$. There is $D = D(\delta, \lambda, C)$ such that any rectifiable σ with* $\mathrm{length}(\sigma|_{[u,v]}) \leq \lambda d(\sigma(u), \sigma(v)) + C$ *for all $u \leq v$ lies in the D-neighborhood of a geodesic J with the same endpoints, and J lies in the $2D$-neighborhood of σ. Triangles whose three sides obey the subarc bound are $(3D + \delta)$-thin.*

Proof. Projection. If $R_0 > 4\delta$, $d(x, J), d(y, J) \geq R_0$, $d(x, y) \leq R_0/2$, then nearest points x', y' satisfy $d(x', y') \leq 8\delta$: every $z \in [x', y']$ is within 2δ of one of the other sides of the quadrilateral x, x', y', y, cannot be near $[x, y]$ (else $d(x, J) \leq R_0/2 + 2\delta < R_0$), and near $[x, x']$ or $[y, y']$ forces $d(z, x') \leq 4\delta$; apply this at the midpoint.

Excursions. Take $R_0 = 1 + 32\lambda\delta$, so $R_0 > 4\delta$ and $16\lambda\delta/R_0 < 1/2$. A component of σ beyond distance R_0 from J has both ends at distance R_0; partition it by arclength into at most $1 + 2H/R_0$ pieces of length $\leq R_0/2$, with H its length. All partition points are at distance $\geq R_0$, so summing the projection bound gives $8\delta(1 + 2H/R_0)$ for the end projections' distance, and the subarc bound gives $H \leq \lambda(2R_0 + 8\delta) + \frac{16\lambda\delta}{R_0} H + C$, hence $H \leq 2[\lambda(2R_0 + 8\delta) + C]$. Every excursion point is within $H/2$ of an end, so σ lies within $D = R_0 + \lambda(2R_0 + 8\delta) + C$ of J. Conversely, for $q \in J$ choose $x \in \sigma$ with $d(s, x) = d(s, q)$ (s the start); a nearest $x' \in J$ gives $d(q, x') = |d(s, x) - d(s, x')| \leq D$, so $d(q, x) \leq 2D$. Replacing path sides by geodesics gives $(3D + \delta)$-thinness. □

The Cayley graph. Word-hyperbolicity is the last step: relate distances in reduced diagrams to word distances in a Cayley graph, then convert diagram thinness into thinness of Cayley triangles.

Proposition 3 (Word-hyperbolic)**.** *$G = \pi_1(K)$ is word-hyperbolic.*

Proof. There is B_1 with $d_{\Delta^{(1)}}(u, v) \leq B_1 d_\Delta(u, v)$ for diagram vertices: replace a straight piece through a triangle interior by a route over its boundary across a corner of angle ω, of length at most $1/\sin(\omega/2)$ times the Euclidean distance (law of cosines), with one factor c over the finitely

many model angles. Partial edge traversals combine into a full-edge walk of length $\le c\, d_\Delta(u,v)$ and at most $c\, d_\Delta(u,v)/\ell_*$ edges, ℓ_* the minimal model edge length.

Fix o and a spanning tree of the 1-skeleton; for an oriented edge $e : u \to v$ take the element of $\tau_u e \tau_v^{-1}$ and let S be the finite symmetric set of nonidentity values, which generates G. Choose one edge loop per $s \in S$ and let B_2 bound all their lengths. Expand a Cayley geodesic triangle $\mathrm{Cay}(G,S)$ letter by letter into these loops. Their concatenation is a nullhomotopic edge path with a reduced diagram; the letter transitions partition each side into loops of length $\le B_2$ whose marks are the successive group vertices.

For marks u, v with labels g_u, g_v, $d_S(g_u, g_v) \le d_{\Delta^{(1)}}(u,v) \le B_1 d_\Delta(u,v)$: any diagram edge path has the same displacement as the outside walk (they differ by a contractible edge loop), and inserting tree paths gives at most one letter of S per edge. Along one geodesic side the letter count between marks equals their group distance, so the expanded subarc between them has length $\le B_1 B_2 d_\Delta(u,v)$; extending an arbitrary subarc to enclosing marks adds at most B_2 per end. With $\lambda = \max\{1, B_1 B_2\}$ and $C = 2\lambda B_2$, every subarc obeys the stability bound. Lemmas 8 and 9 give thinness $T = 3D + \delta$ of the expanded path triangle; a mark on a side is within $T + B_2$ of a mark on another side, so group vertices are thin with $H_0 = B_1(T + B_2)$, $H = H_0 + 1$. Replacing vertexless sides and tails by segments at distance ≤ 1, and reducing arbitrary geodesic triangles to vertex polygons, finally gives a uniform thinness constant $4H + 2$ for all geodesic triangles in $\mathrm{Cay}(G,S)$, which is word-hyperbolicity. $\square$

The closed-path statement, torsion-freeness, and word-hyperbolicity together prove the angular criterion, hence the main theorem.